

1. Frequency Tables

[DataSet1]

		Statistics			
		Age Categories	Preferred_Payment_Method	App_Usage_Frequency	Income_Level
N	Valid	641	641	641	641
	Missing	0	0	0	0

Age Group Distribution

- Most participants belong to the **Gen Z (16–25)** and **Millennials (26–40)** categories, showing that the sample is concentrated in younger age groups.
- **Baby Boomers (56–69)** have the smallest representation, which is reasonable in a study focused on mobile wallet usage.
- Overall, the sample is suitable for examining digital payment behavior across different age groups

		Age Categories			
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	1.00	113	17.6	17.6	17.6
	2.00	161	25.1	25.1	42.7
	3.00	202	31.5	31.5	74.3
	4.00	165	25.7	25.7	100.0
	Total	641	100.0	100.0	

Preferred Payment Method

- **UPI** is the most commonly preferred payment option, followed by **Wallet Balance**. This indicates strong acceptance of digital and mobile wallet methods.
- **Debit Card** and **Credit Card** are less frequently chosen, suggesting lower preference for traditional payment methods.

Preferred_Payment_Method

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Credit Card	161	25.1	25.1	25.1
	Debit Card	157	24.5	24.5	49.6
	UPI	159	24.8	24.8	74.4
	Wallet Balance	164	25.6	25.6	100.0
	Total	641	100.0	100.0	

App Usage Frequency

- Most respondents reported using the application **daily**, while fewer users selected weekly or monthly usage.
- This reflects a high level of engagement with the payment app, especially among younger individuals.

App_Usage_Frequency

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Daily	229	35.7	35.7	35.7
	Monthly	203	31.7	31.7	67.4
	Weekly	209	32.6	32.6	100.0
	Total	641	100.0	100.0	

Income Level

- The majority of respondents fall within the **middle-income category**, followed by the low-income group.
- High-income respondents are comparatively fewer, which may affect spending levels but may not strongly impact usage frequency.

		Income_Level			
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	High	202	31.5	31.5	31.5
	Low	219	34.2	34.2	65.7
	Middle	220	34.3	34.3	100.0
	Total	641	100.0	100.0	

2. Measures of Central Tendency and Dispersion

Age

- The mean age is slightly higher than the median, indicating a mild positive skewness.
- Standard deviation shows a moderate spread in ages.
- The range confirms participation from both younger and older individuals.

Total Transactions

- Since the mean exceeds the median, the variable appears positively skewed, meaning some users make transactions very frequently.
- A high variance suggests considerable differences in transaction behavior among respondents.

Total Spent

- A wide range and large standard deviation indicate that while many users spend moderate amounts, some spend significantly more.

LTV (Lifetime Value)

- The mean is higher than the median, showing positive skewness.
- A few high-value users increase the overall average.

Statistics								
		Age	Total_Transactions	Avg_Transaction_Value	Total_Spent	Active_Days	LTV	Cashback_Received
N	Valid	641	641	641	641	641	641	641
	Missing	0	0	0	0	0	0	0
Mean		43.27	513.77	9459.502626	4802949.589	182.99	491860.9313	2467.849248
Median		44.00	519.00	9392.947191	3451735.371	185.00	358998.9063	2438.786722
Mode		48	463 ^a	35.1233399 ^a	4117.38666 ^a	5 ^a	6300.97867 ^a	.234348958 ^a
Std. Deviation		15.548	285.295	5715.873254	4219835.773	106.529	422237.8085	1435.860687
Variance		241.734	81393.268	32671207.05	1.781E+13	11348.426	1.783E+11	2061695.911
Range		53	998	19939.35452	18271162.27	364	1835899.901	4999.464131
Minimum		16	2	35.12333992	4117.386657	1	6300.978674	.2343489580
Maximum		69	1000	19974.47786	18275279.66	365	1842200.880	4999.698480

a. Multiple modes exist. The smallest value is shown

3. Cross-Tabulation

Age Group × Preferred Payment Method

- **Gen Z** and **Millennials** mainly prefer **UPI** and **Wallet Balance**.
- **Gen X** and **Baby Boomers** show relatively greater use of **Debit Cards** and **Credit Cards**.
- This suggests that younger groups are more likely to adopt digital wallet services.

Case Processing Summary						
		Cases				
		Valid		Missing		Total
		N	Percent	N	Percent	N Percent
Age Categories * Preferred_Payment_Method		641	100.0%	0	0.0%	641 100.0%

Age Categories * Preferred_Payment_Method Crosstabulation							
			Preferred_Payment_Method				
			Credit Card	Debit Card	UPI	Wallet Balance	Total
Age Categories	1.00	Count	25	26	34	28	113
		% within Age Categories	22.1%	23.0%	30.1%	24.8%	100.0%
		% within Preferred_Payment_Meth od	15.5%	16.6%	21.4%	17.1%	17.6%
	2.00	Count	45	32	44	40	161
		% within Age Categories	28.0%	19.9%	27.3%	24.8%	100.0%
		% within Preferred_Payment_Meth od	28.0%	20.4%	27.7%	24.4%	25.1%
	3.00	Count	51	54	40	57	202
		% within Age Categories	25.2%	26.7%	19.8%	28.2%	100.0%
		% within Preferred_Payment_Meth od	31.7%	34.4%	25.2%	34.8%	31.5%
	4.00	Count	40	45	41	39	165
		% within Age Categories	24.2%	27.3%	24.8%	23.6%	100.0%
		% within Preferred_Payment_Meth od	24.8%	28.7%	25.8%	23.8%	25.7%
Total	Count	161	157	159	164	641	
	% within Age Categories	25.1%	24.5%	24.8%	25.6%	100.0%	
	% within Preferred_Payment_Meth od	100.0%	100.0%	100.0%	100.0%	100.0%	

Age Group × App Usage Frequency

- Daily app usage is highest among **Gen Z** and gradually declines with older age groups.
- **Baby Boomers** have the highest share of monthly users.
- This indicates that app usage frequency decreases as age increases.

Case Processing Summary						
	Valid		Cases Missing		Total	
	N	Percent	N	Percent	N	Percent
Age Categories * App_Usage_Frequency	641	100.0%	0	0.0%	641	100.0%

Age Categories * App_Usage_Frequency Crosstabulation

			App_Usage_Frequency			Total
			Daily	Monthly	Weekly	
Age Categories	1.00	Count	37	35	41	113
		% within Age Categories	32.7%	31.0%	36.3%	100.0%
		% within App_Usage_Frequency	16.2%	17.2%	19.6%	17.6%
	2.00	Count	61	50	50	161
		% within Age Categories	37.9%	31.1%	31.1%	100.0%
		% within App_Usage_Frequency	26.6%	24.6%	23.9%	25.1%
	3.00	Count	68	66	68	202
		% within Age Categories	33.7%	32.7%	33.7%	100.0%
		% within App_Usage_Frequency	29.7%	32.5%	32.5%	31.5%
	4.00	Count	63	52	50	165
		% within Age Categories	38.2%	31.5%	30.3%	100.0%
		% within App_Usage_Frequency	27.5%	25.6%	23.9%	25.7%
Total	Count		229	203	209	641
	% within Age Categories		35.7%	31.7%	32.6%	100.0%
	% within App_Usage_Frequency		100.0%	100.0%	100.0%	100.0%

Age Group × Income Level

- Income levels generally rise with older age groups.
- However, younger lower-income groups still show strong digital payment preference, implying that age has a stronger effect than income.

Case Processing Summary						
	Cases					
	Valid		Missing		Total	
	N	Percent	N	Percent	N	Percent
Age Categories * Income_Level	641	100.0%	0	0.0%	641	100.0%

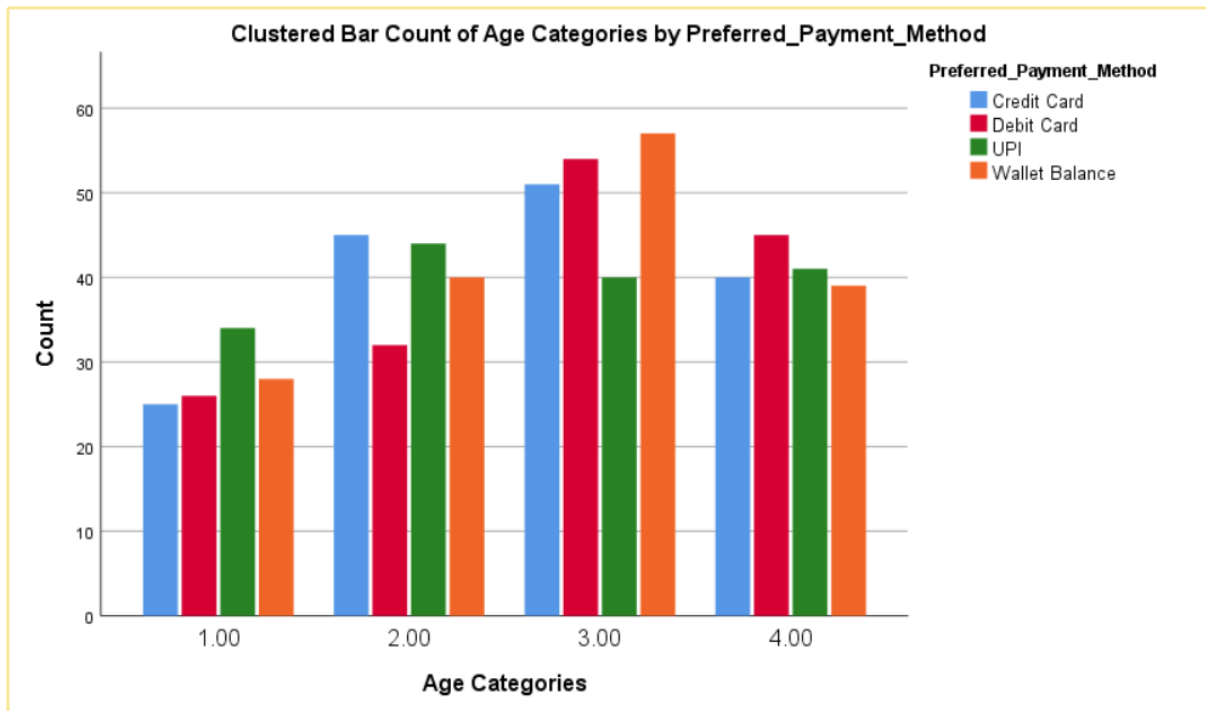
Age Categories * Income_Level Crosstabulation

			Income_Level			
			High	Low	Middle	Total
Age Categories	1.00	Count	31	42	40	113
		% within Age Categories	27.4%	37.2%	35.4%	100.0%
		% within Income_Level	15.3%	19.2%	18.2%	17.6%
	2.00	Count	57	56	48	161
		% within Age Categories	35.4%	34.8%	29.8%	100.0%
		% within Income_Level	28.2%	25.6%	21.8%	25.1%
	3.00	Count	62	69	71	202
		% within Age Categories	30.7%	34.2%	35.1%	100.0%
		% within Income_Level	30.7%	31.5%	32.3%	31.5%
	4.00	Count	52	52	61	165
		% within Age Categories	31.5%	31.5%	37.0%	100.0%
		% within Income_Level	25.7%	23.7%	27.7%	25.7%
Total	Count	202	219	220	641	
	% within Age Categories	31.5%	34.2%	34.3%	100.0%	
	% within Income_Level	100.0%	100.0%	100.0%	100.0%	

4. Graphical Analysis

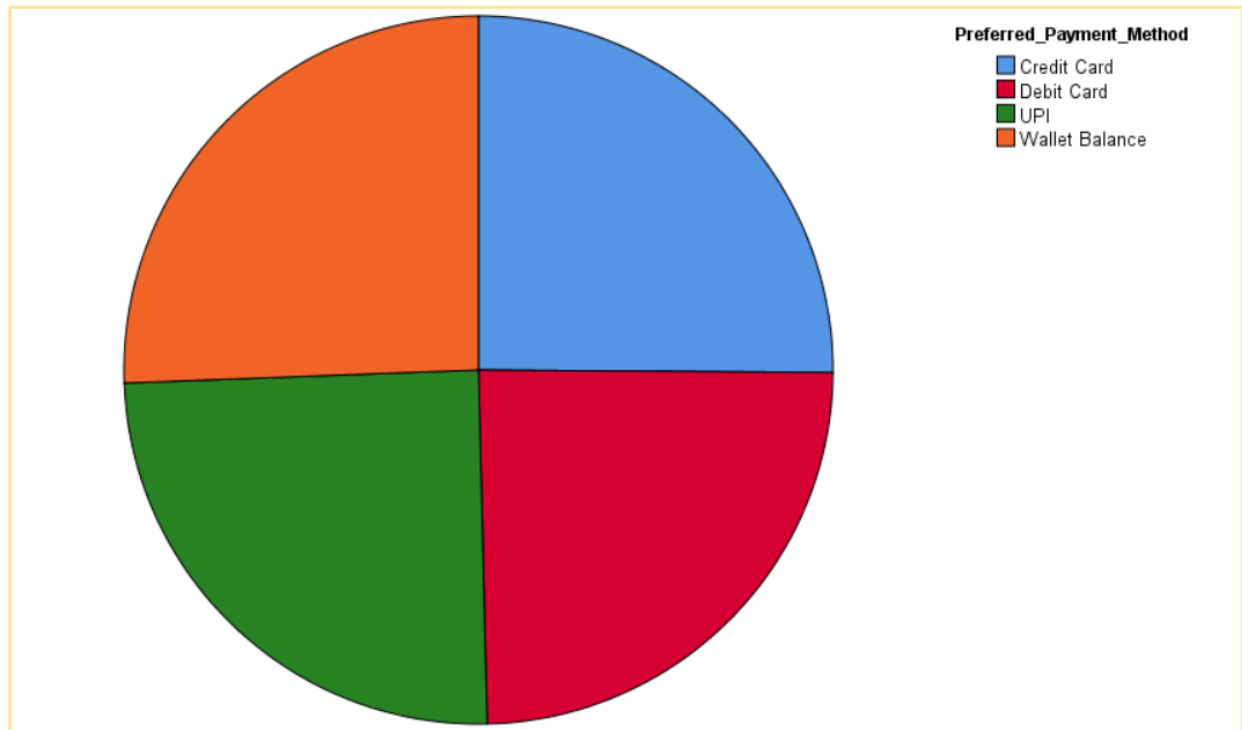
Bar Chart: Age Group vs Preferred Payment Method

- The chart shows that **UPI** is the leading payment method among younger respondents.
- Wallet Balance is also widely used among Gen Z and Millennials.
- Credit card usage becomes slightly more common in older groups.



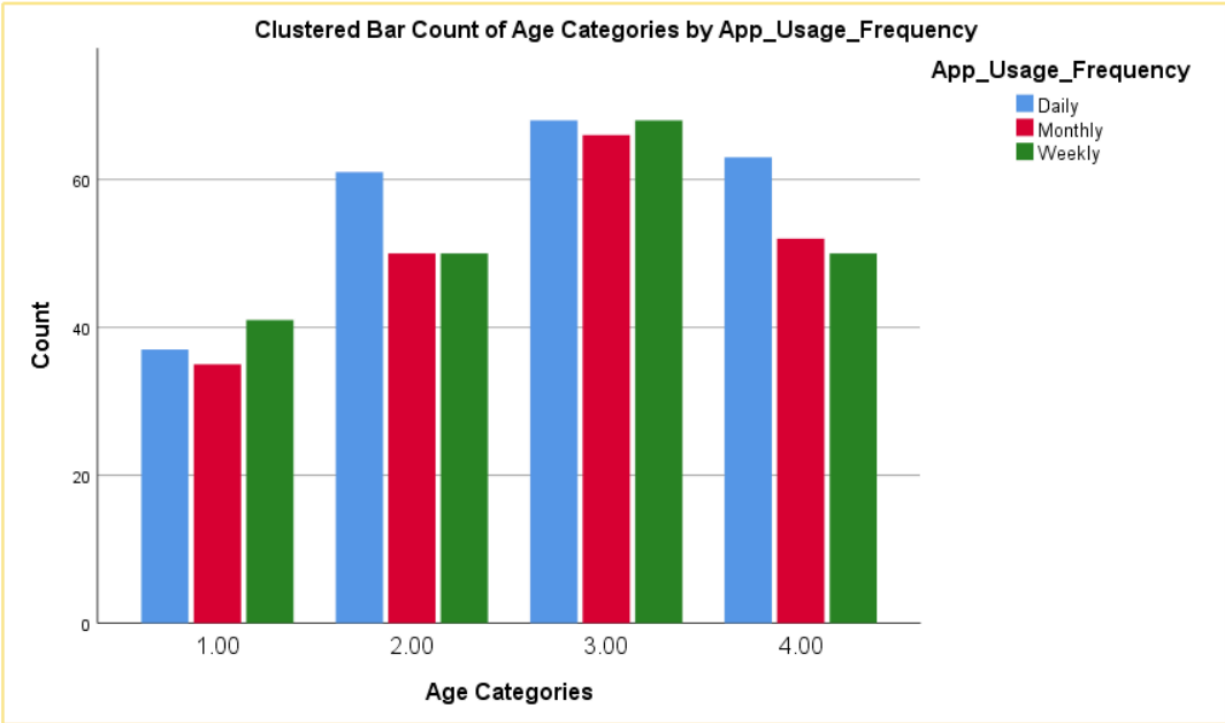
Pie Chart: Preferred Payment Method

- **UPI** accounts for the largest share of payment preferences.
- Combined with Wallet Balance, digital methods clearly dominate the sample.



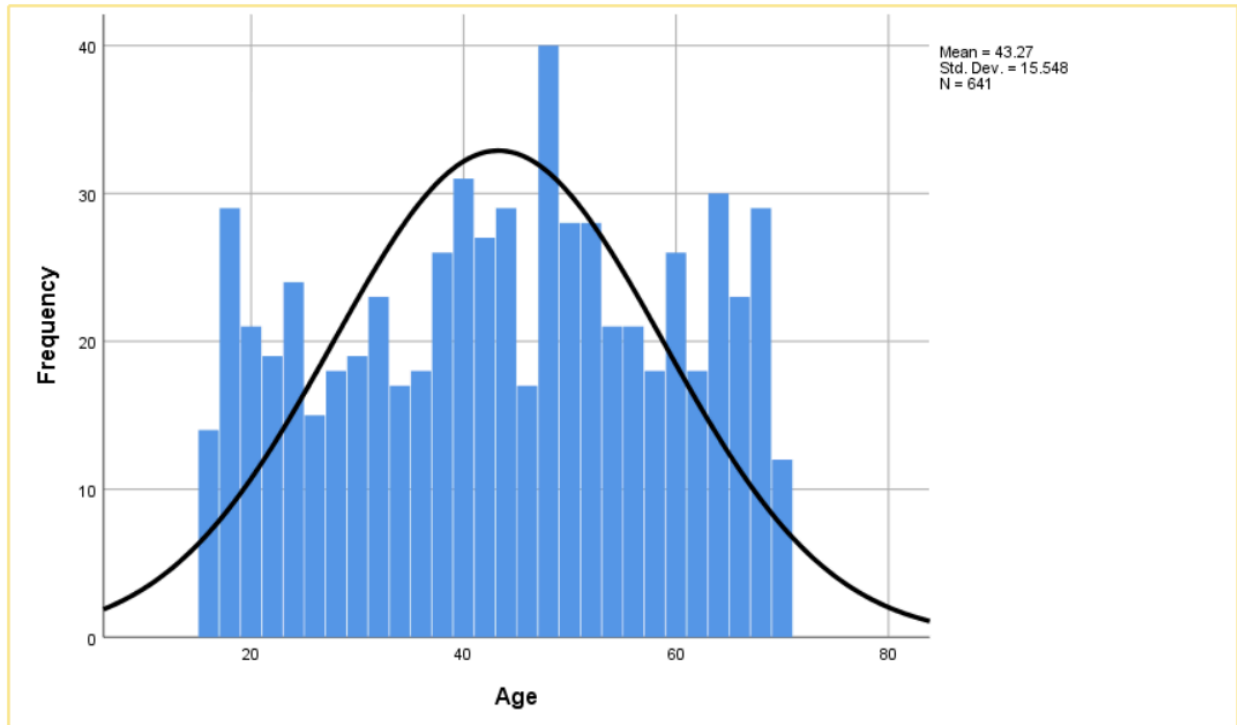
Bar Chart: App Usage Frequency by Age Group

- Younger groups use the app more frequently than older groups.
- Gen Z users are mostly daily users, while Baby Boomers are more likely monthly users.



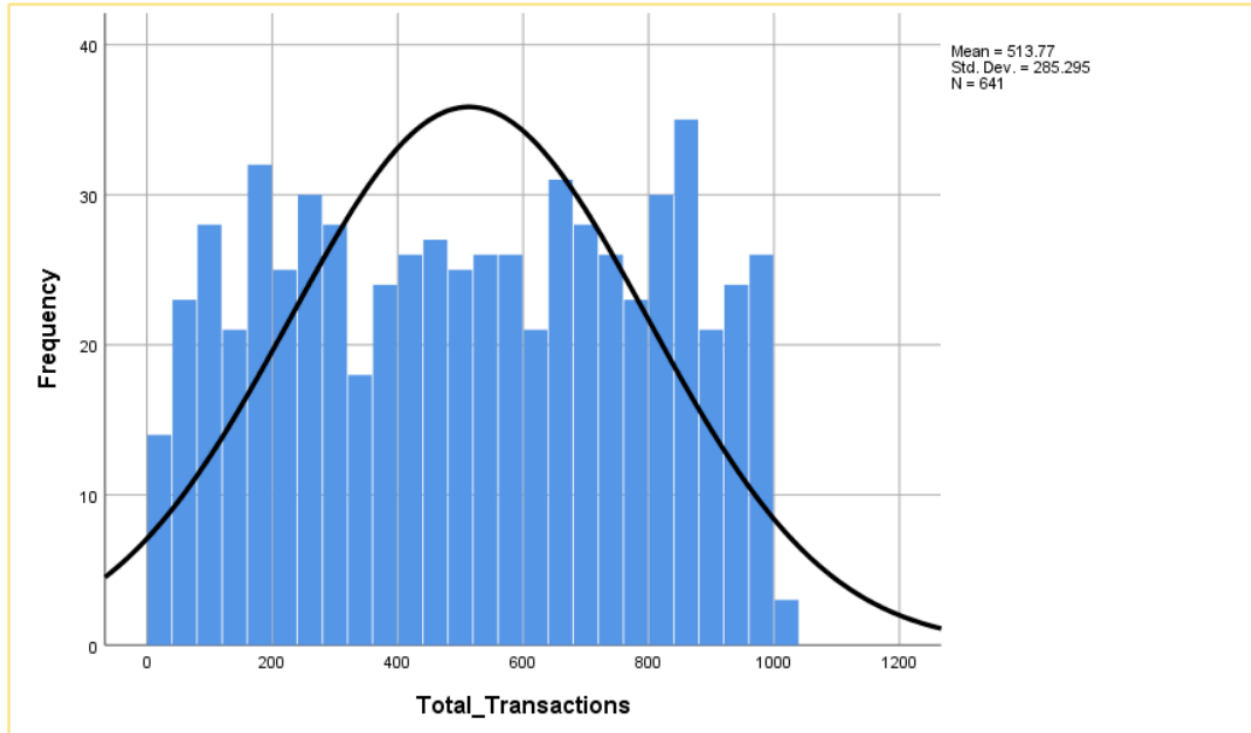
Histogram: Age

- The age distribution is slightly positively skewed, with most respondents between 20 and 40 years.



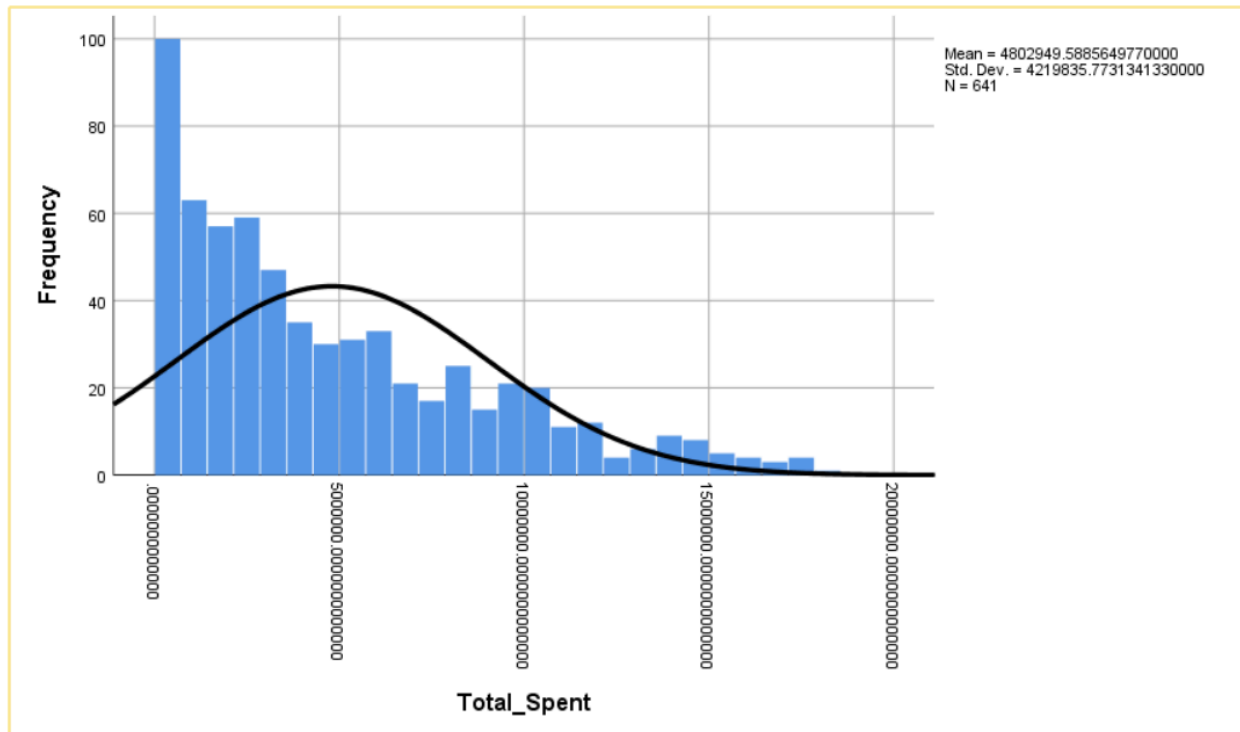
Histogram: Total Transactions

- The data is positively skewed, showing that most users make low to moderate transactions while a few transact heavily.



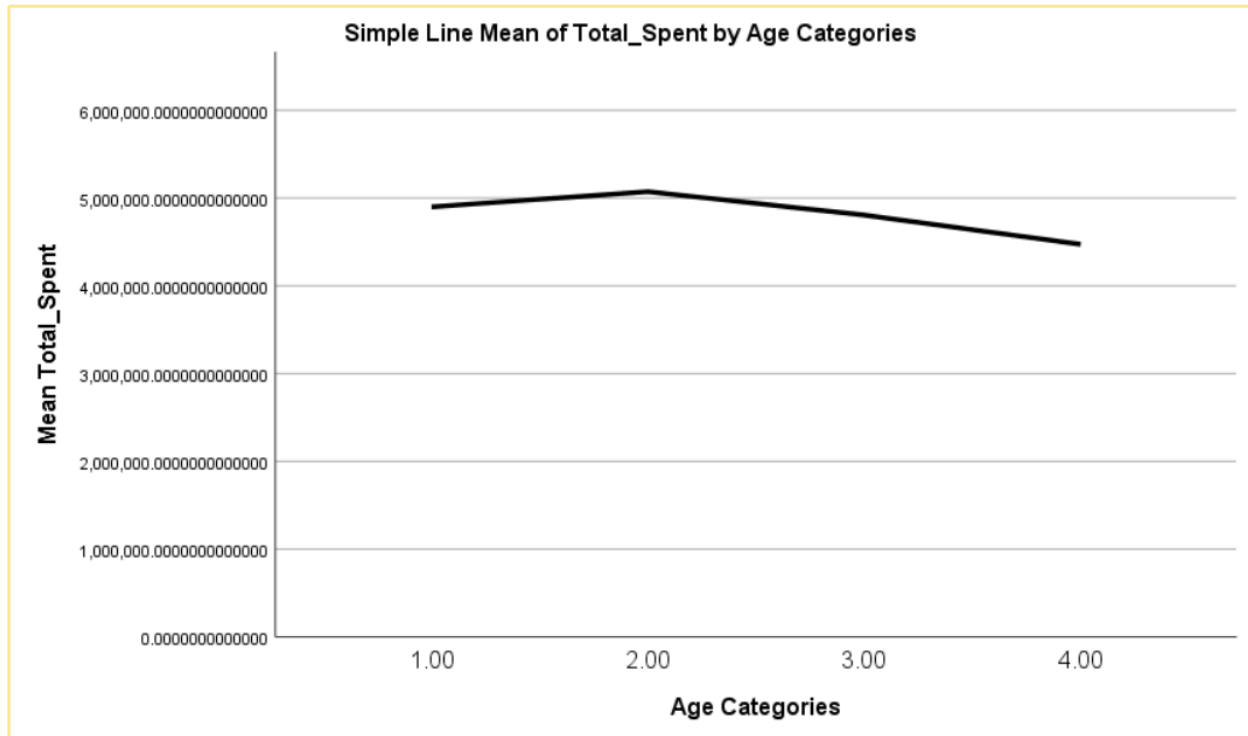
Histogram: Total Spent

- The distribution is highly positively skewed, with most users spending smaller amounts and a few spending very high amounts.



Line Graph: Average Total Spent across Age Groups

- Gen Z and Millennials show the highest average spending.
- This suggests that familiarity with digital payments may influence spending more than income alone.



5. Distribution Shape and Normality

Skewness and Kurtosis

- **Age** shows slight positive skewness and is close to a normal shape.
- **Total Transactions** has strong positive skewness and is not normally distributed.
- **Total Spent** is highly positively skewed and deviates from normality.
- **LTV** also shows positive skewness.

Normality Tests (Shapiro-Wilk / Kolmogorov-Smirnov)

- All significance values are below 0.05, indicating that the continuous variables are not normally distributed.
- Such results are common in financial and transaction-based datasets.

Interpretation

- Although some parametric tests assume normality, methods such as ANOVA are often robust with larger samples.
- However, non-parametric alternatives like Kruskal-Wallis may also be considered for more accurate results.

Case Processing Summary

	Valid		Cases Missing		Total	
	N	Percent	N	Percent	N	Percent
Age	641	100.0%	0	0.0%	641	100.0%
Total_Transactions	641	100.0%	0	0.0%	641	100.0%
Total_Spent	641	100.0%	0	0.0%	641	100.0%
LTV	641	100.0%	0	0.0%	641	100.0%

Descriptives

			Statistic	Std. Error
Age	Mean		43.27	.614
	95% Confidence Interval for Mean	Lower Bound	42.07	
		Upper Bound	44.48	
	5% Trimmed Mean		43.37	
	Median		44.00	
	Variance		241.734	
	Std. Deviation		15.548	
	Minimum		16	
	Maximum		69	
	Range		53	
	Interquartile Range		25	
	Skewness		-.102	.097
	Kurtosis		-1.103	.193
Total_Transactions	Mean		513.77	11.268
	95% Confidence Interval for Mean	Lower Bound	491.64	
		Upper Bound	535.90	
	5% Trimmed Mean		514.19	
	Median		519.00	
	Variance		81393.268	

	Variance		61393.206	
	Std. Deviation		285.295	
	Minimum		2	
	Maximum		1000	
	Range		998	
	Interquartile Range		501	
	Skewness		-.025	.097
	Kurtosis		-1.213	.193
Total_Spent	Mean		4802949.589	166673.4921
	95% Confidence Interval for Mean	Lower Bound	4475656.592	
		Upper Bound	5130242.585	
	5% Trimmed Mean		4466982.484	
	Median		3451735.371	
	Variance		1.781E+13	
	Std. Deviation		4219835.773	
	Minimum		4117.386657	
	Maximum		18275279.66	
	Range		18271162.27	
	Interquartile Range		5950638.631	
	Skewness		1.023	.097
	Kurtosis		.320	.193
LTV	Mean		491860.9313	16677.39074
	95% Confidence Interval for Mean	Lower Bound	459111.9133	
		Upper Bound	524609.9492	
	5% Trimmed Mean		458270.2508	
	Median		358998.9063	
	Variance		1.783E+11	
	Std. Deviation		422237.8085	
	Minimum		6300.978674	
	Maximum		1842200.880	
	Range		1835899.901	
	Interquartile Range		596047.8261	

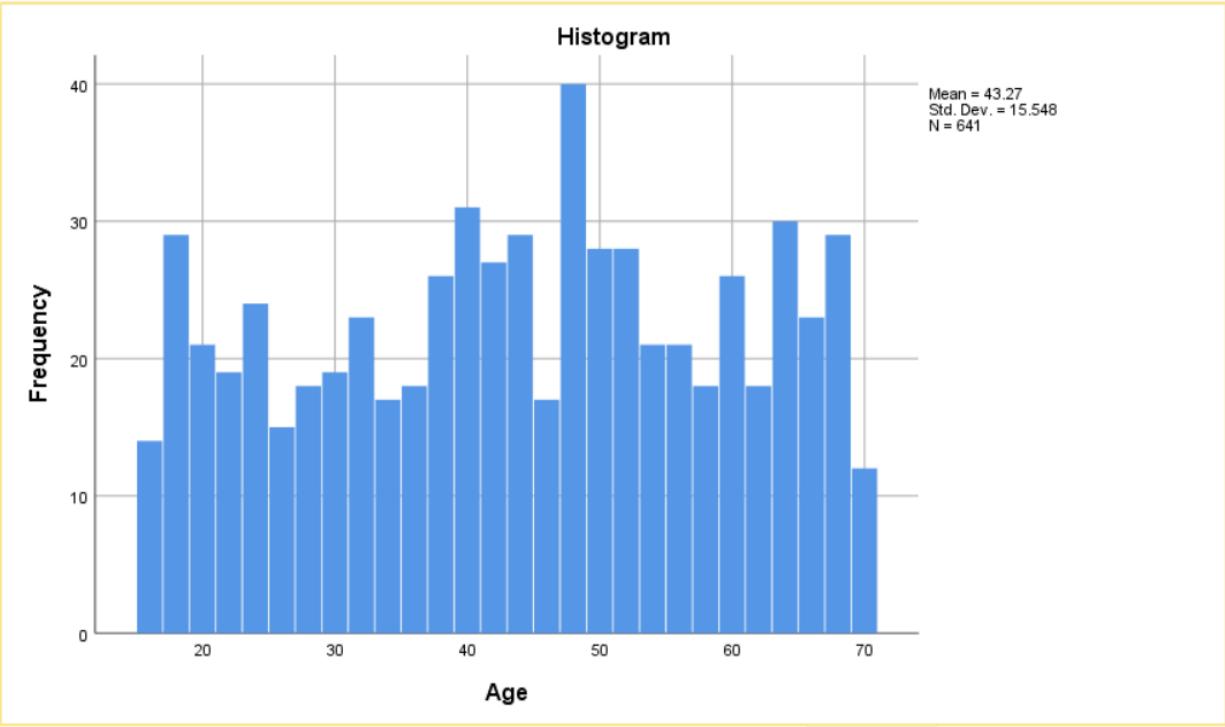
	Interquartile Range	596047.8261	
	Skewness	1.022	.097
	Kurtosis	.317	.193

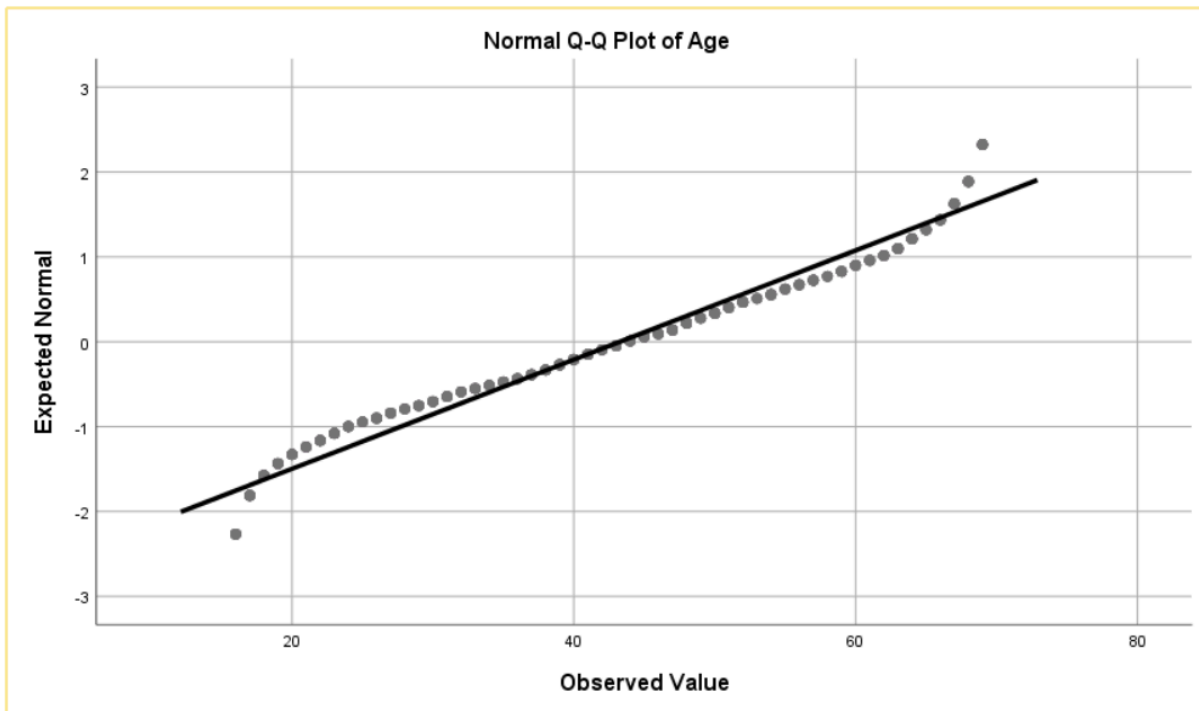
Tests of Normality						
	Kolmogorov-Smirnov ^a			Shapiro-Wilk		
	Statistic	df	Sig.	Statistic	df	Sig.
Age	.061	641	.000	.957	641	.000
Total_Transactions	.071	641	.000	.954	641	.000
Total_Spent	.130	641	.000	.893	641	.000
LTV	.130	641	.000	.894	641	.000

a. Lilliefors Significance Correction

Age:

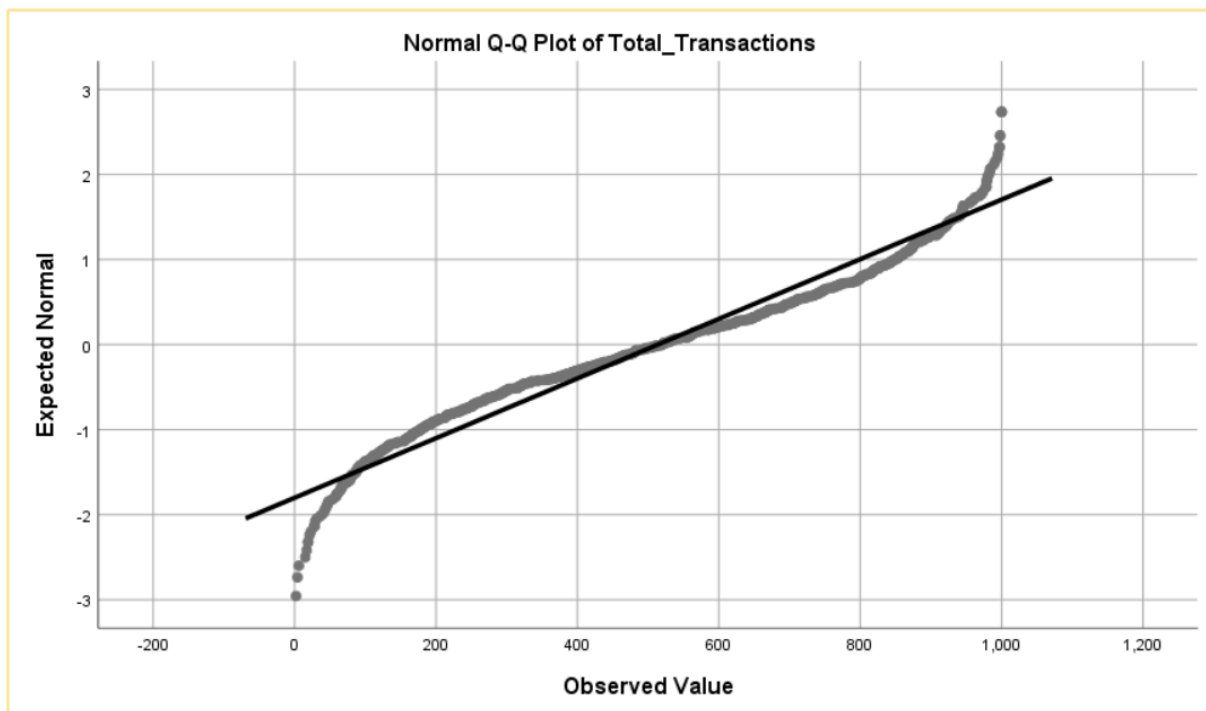
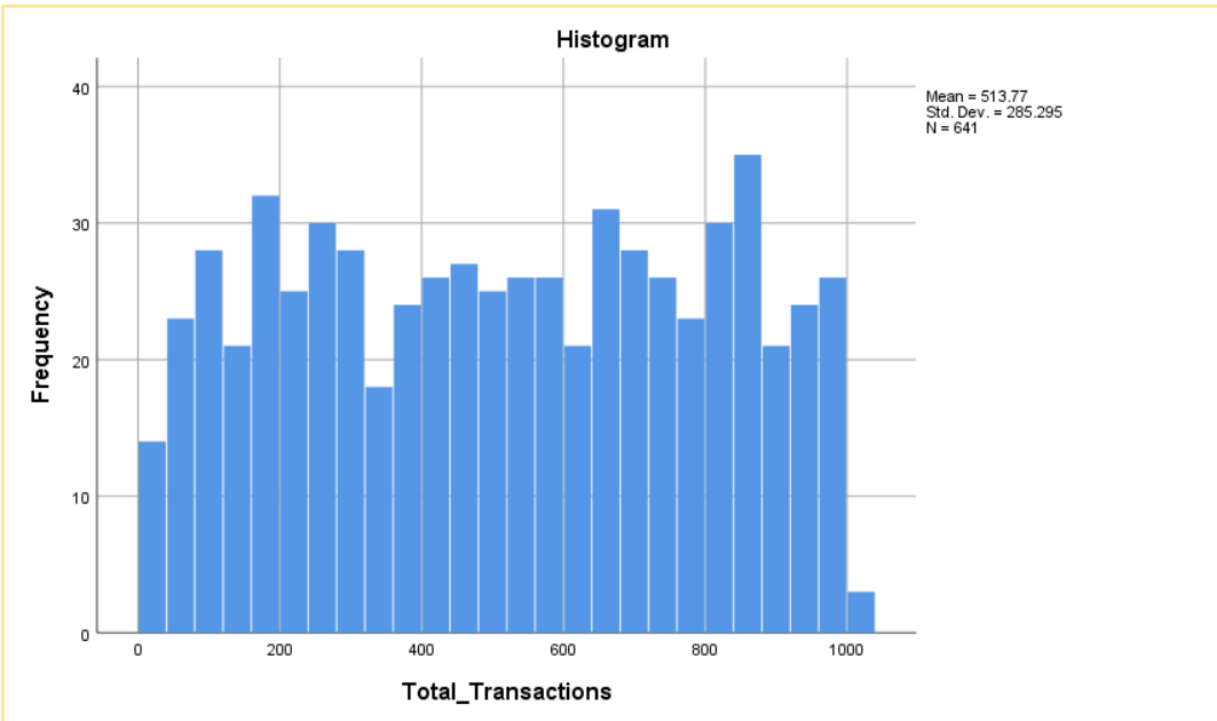
Age

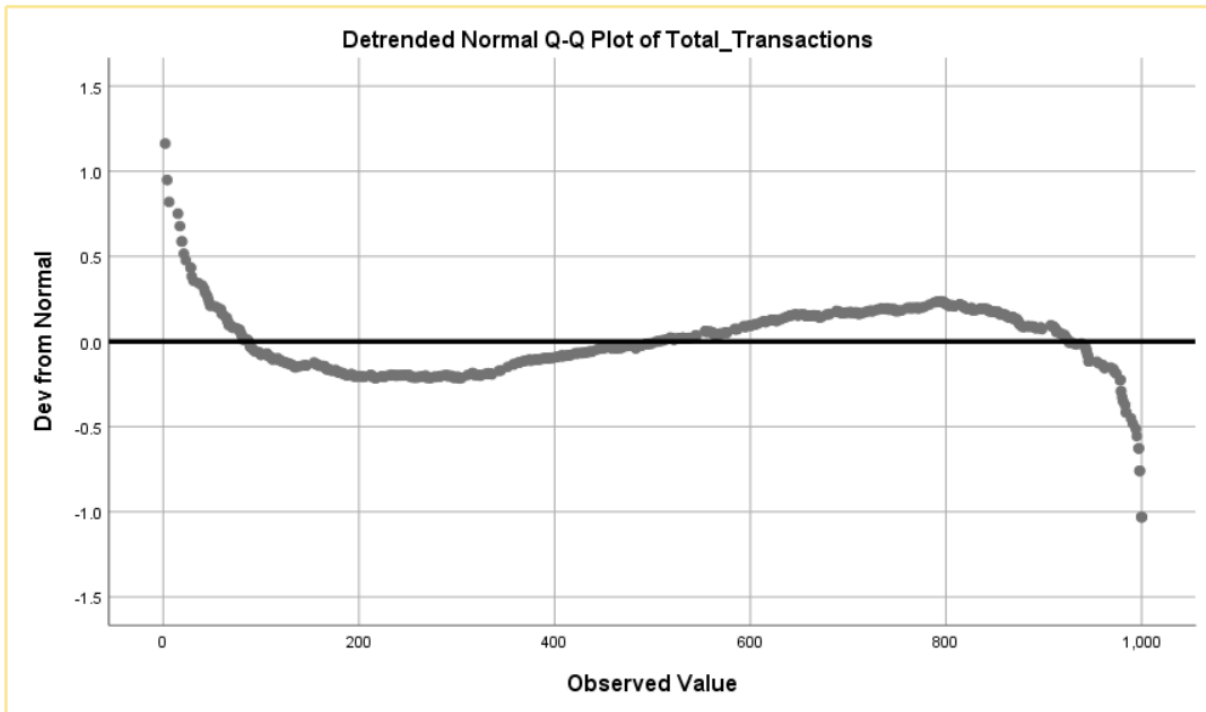




Total Transactions:

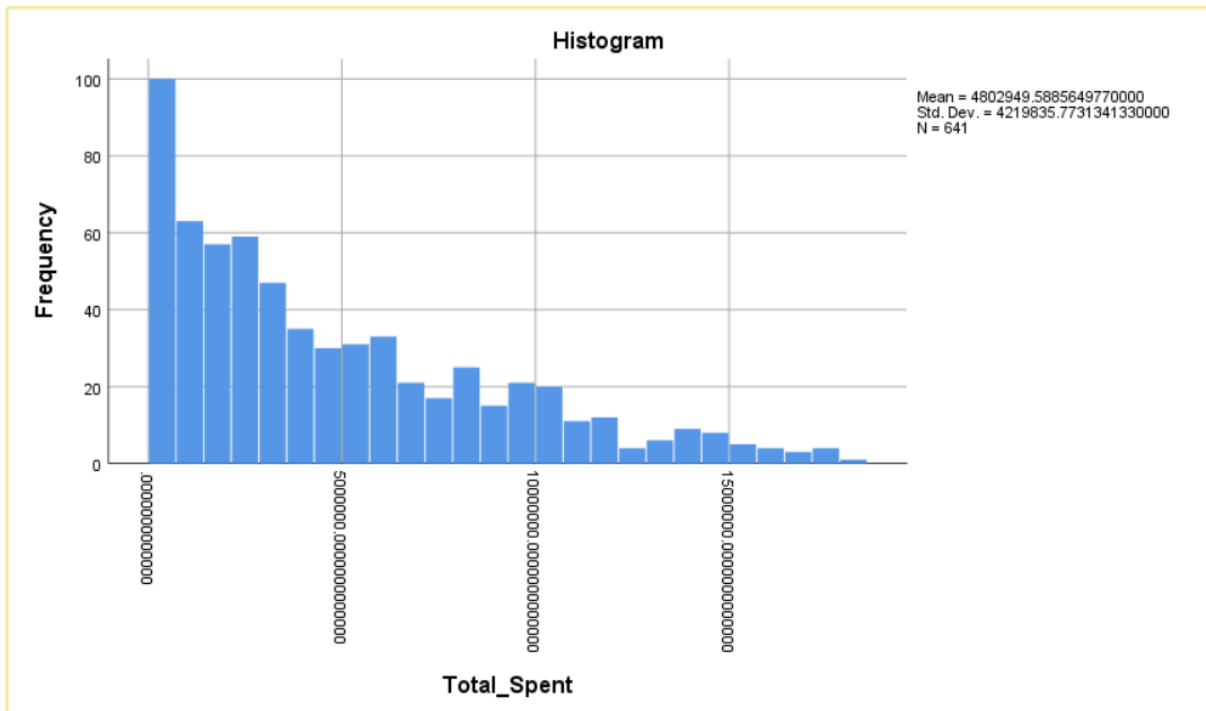
Total_Transactions

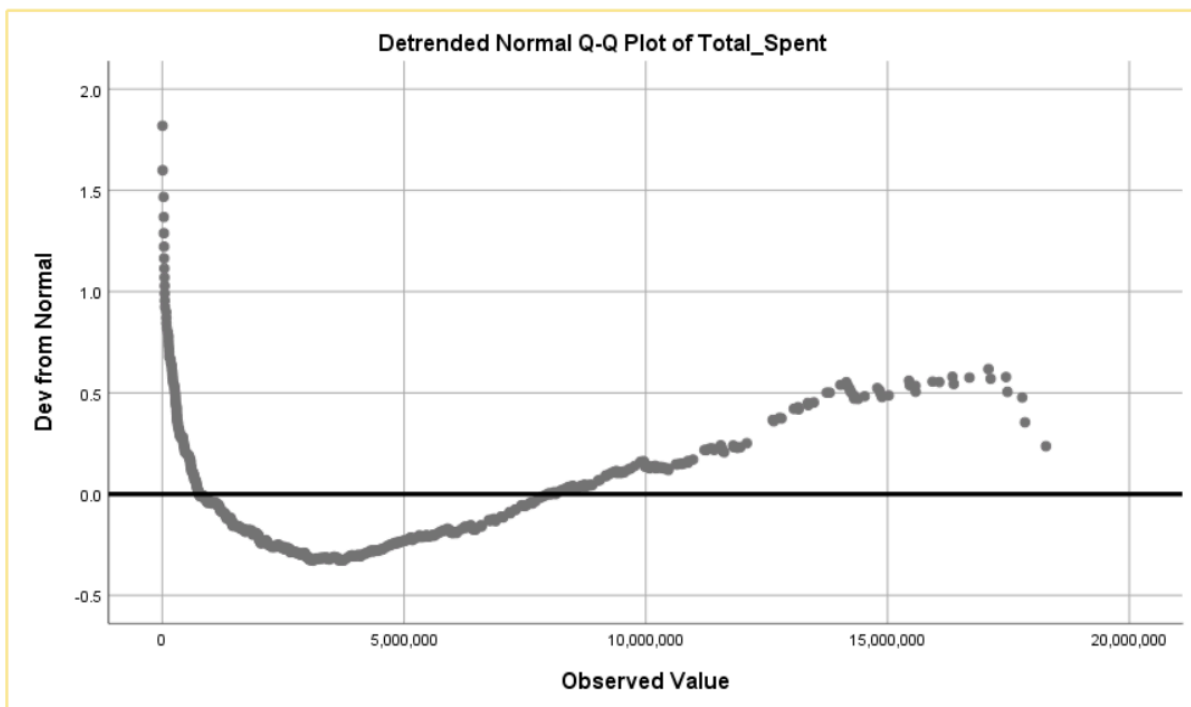
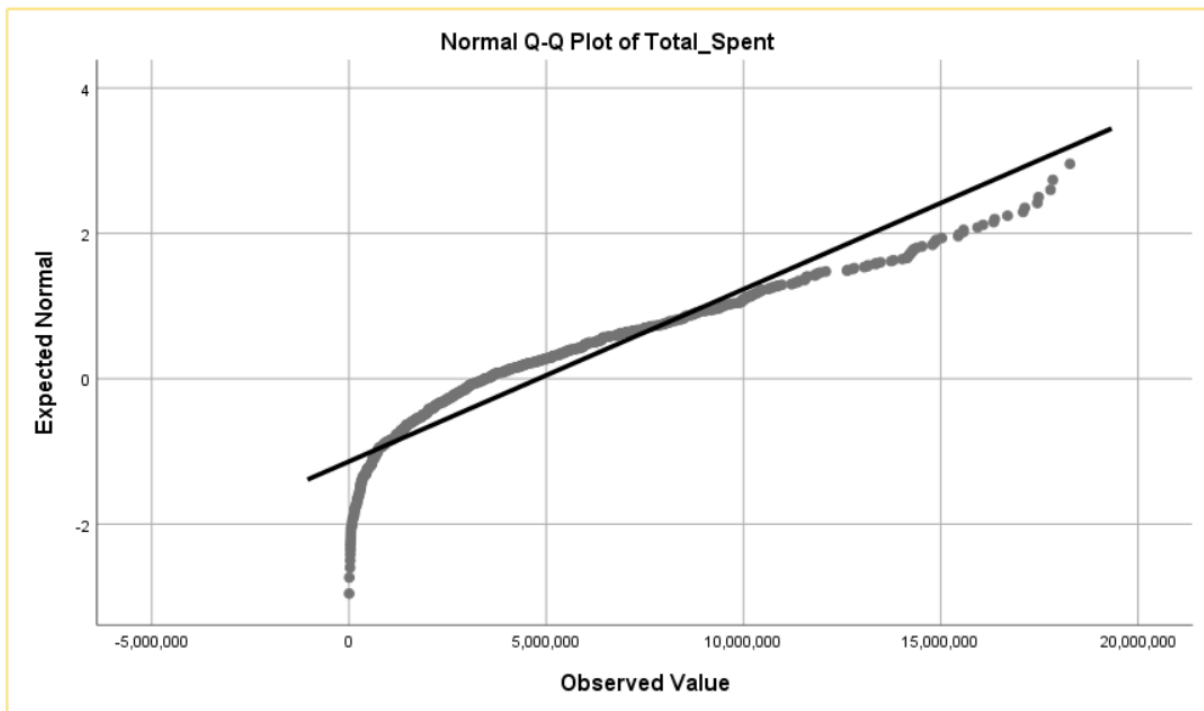




Total Spent:

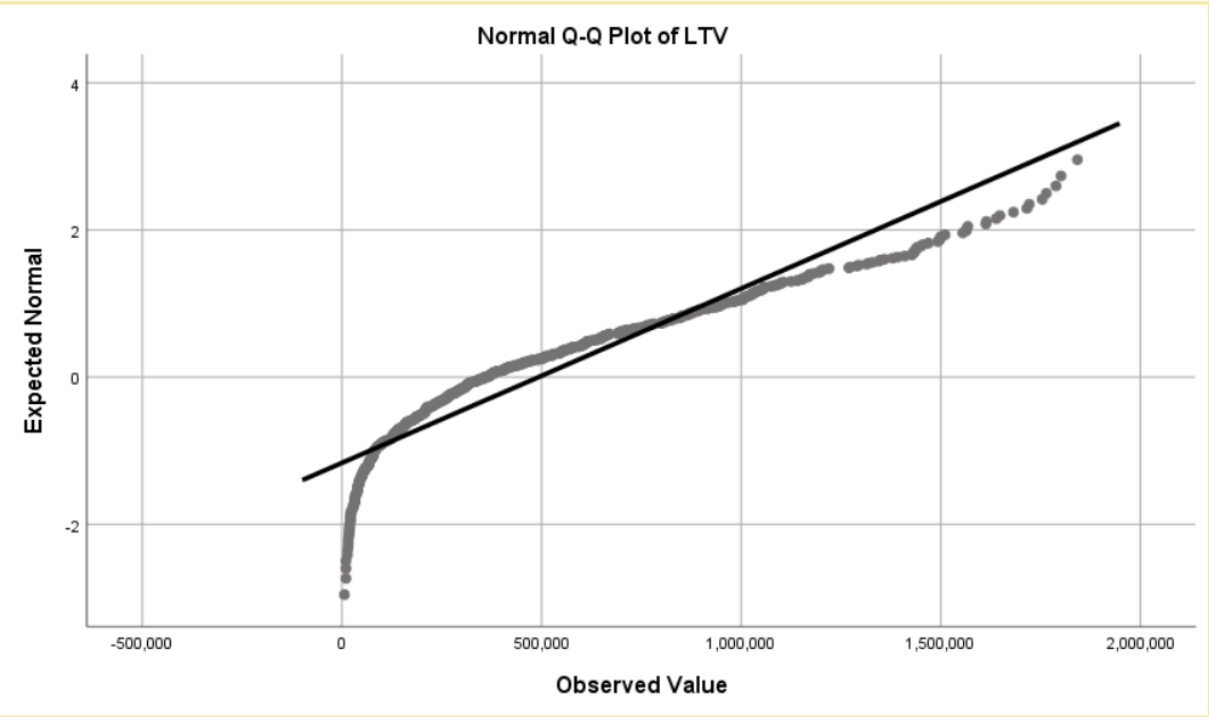
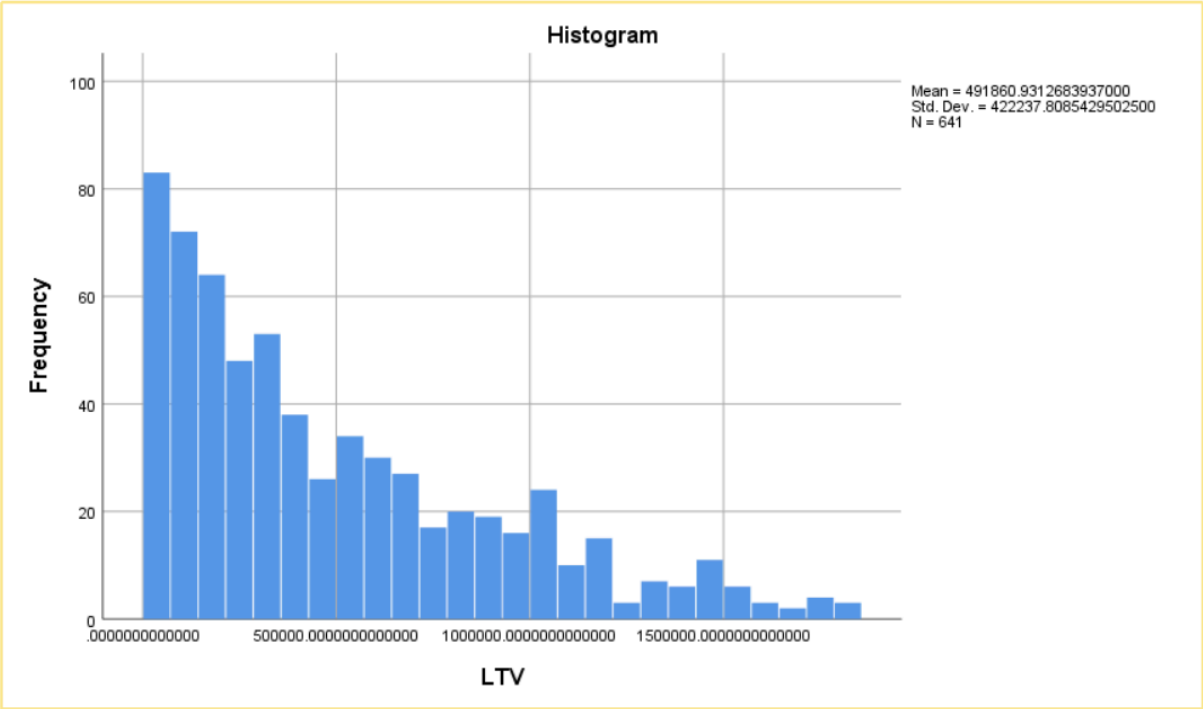
Total_Spent

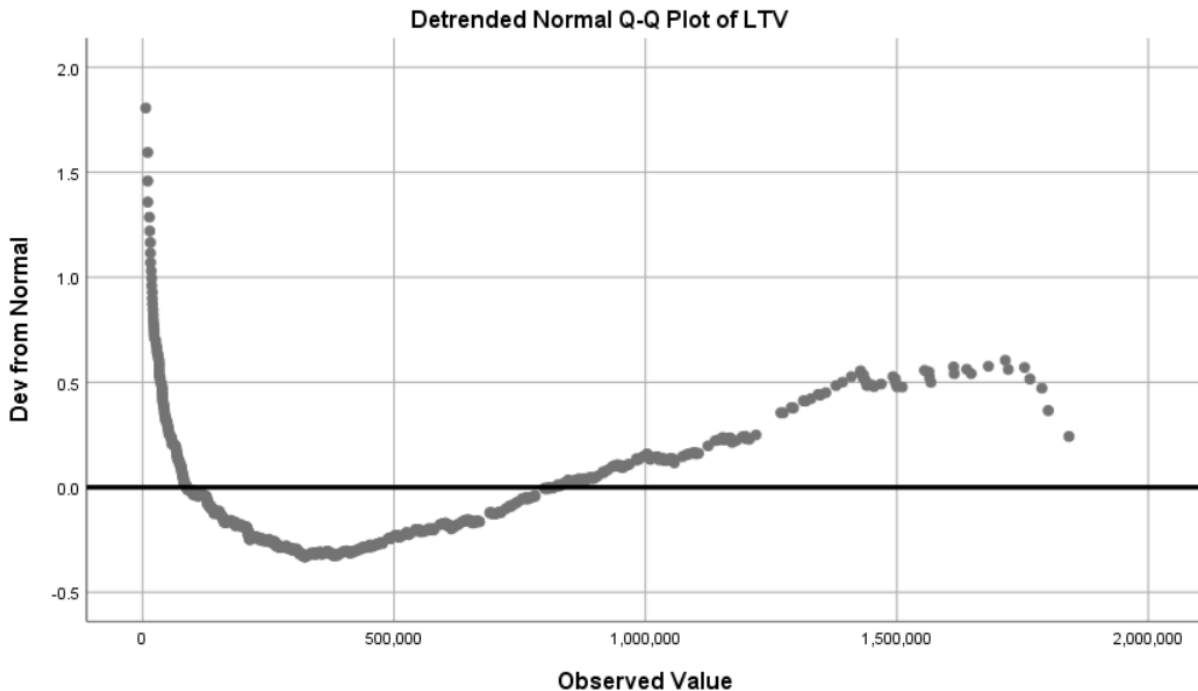




LTV:

LTV





6. Outlier Detection

Boxplots

- **Total Transactions** contains several high-end outliers representing very frequent users.
- **Total Spent** includes extreme outliers, where a few users spent much more than others.
- **Average Transaction Value** has moderate outliers on the higher side.
- **LTV** also contains extreme outliers caused by highly valuable users.

Interpretation

- Many outliers seem to belong to younger groups such as Gen Z and Millennials.
- These may represent heavy or loyal users rather than errors in data.
- For further analysis, results can be compared with and without outliers to check consistency.

Explore

Case Processing Summary						
	Valid		Cases Missing		Total	
	N	Percent	N	Percent	N	Percent
Total_Transactions	641	100.0%	0	0.0%	641	100.0%
Total_Spent	641	100.0%	0	0.0%	641	100.0%
Avg_Transaction_Value	641	100.0%	0	0.0%	641	100.0%
LTV	641	100.0%	0	0.0%	641	100.0%

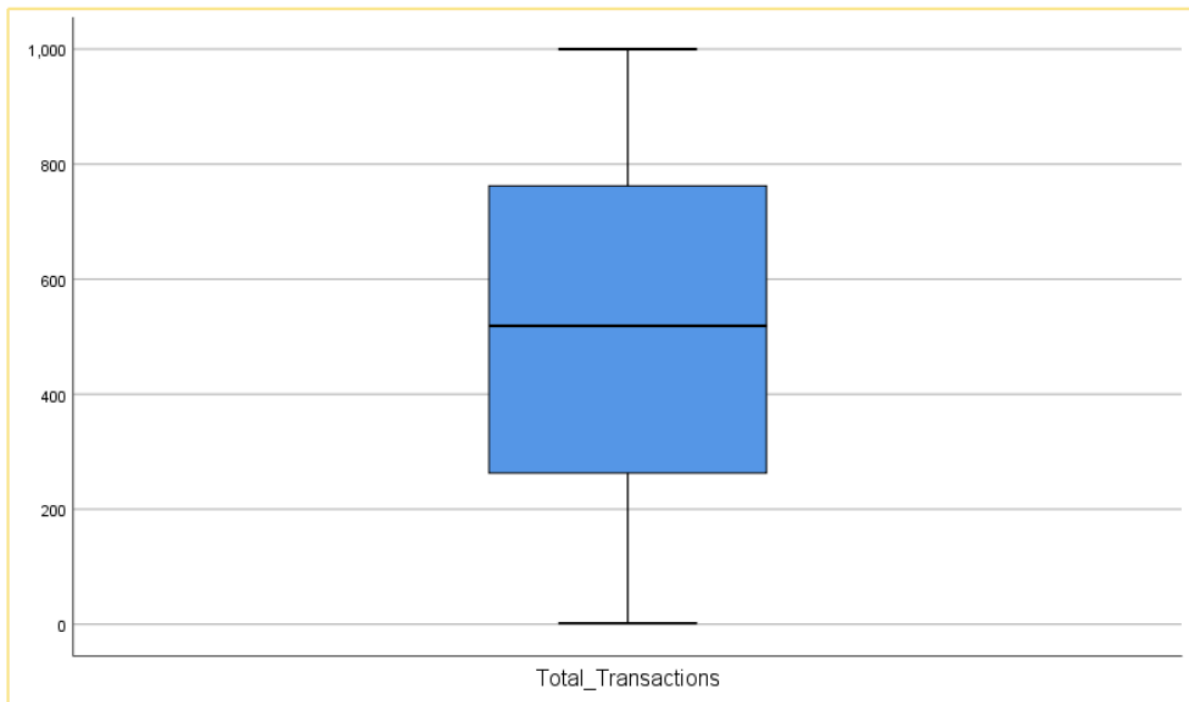
Descriptives

			Statistic	Std. Error
Total_Transactions	Mean		513.77	11.268
	95% Confidence Interval for Mean	Lower Bound	491.64	
		Upper Bound	535.90	
	5% Trimmed Mean		514.19	
	Median		519.00	
	Variance		81393.268	
	Std. Deviation		285.295	
	Minimum		2	
	Maximum		1000	
	Range		998	
	Interquartile Range		501	
	Skewness		-.025	.097
	Kurtosis		-1.213	.193
Total_Spent	Mean		4802949.589	166673.4921
	95% Confidence Interval for Mean	Lower Bound	4475656.592	
		Upper Bound	5130242.585	
	5% Trimmed Mean		4466982.484	

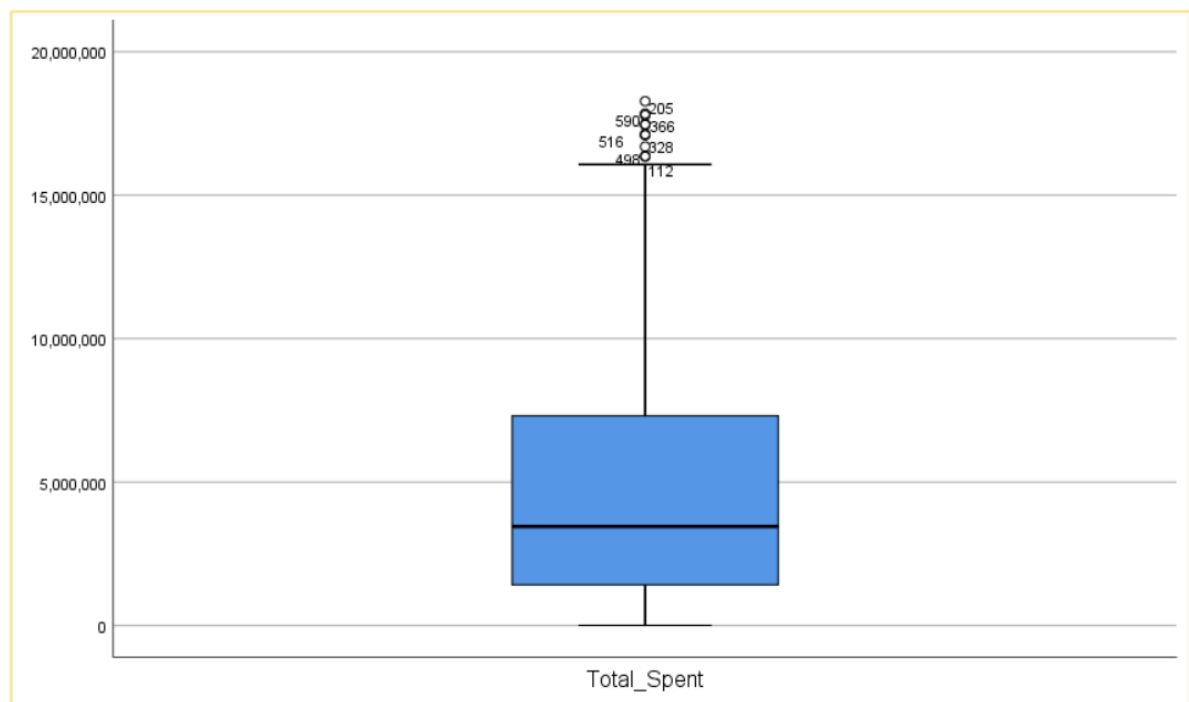
	5% Trimmed Mean		4466982.484	
	Median		3451735.371	
	Variance		1.781E+13	
	Std. Deviation		4219835.773	
	Minimum		4117.386657	
	Maximum		18275279.66	
	Range		18271162.27	
	Interquartile Range		5950638.631	
	Skewness		1.023	.097
	Kurtosis		.320	.193
Avg_Transaction_Value	Mean		9459.502626	225.7634199
	95% Confidence Interval for Mean	Lower Bound	9016.176065	
		Upper Bound	9902.829187	
	5% Trimmed Mean		9406.523091	
	Median		9392.947191	
	Variance		32671207.05	
	Std. Deviation		5715.873254	
	Minimum		35.12333992	
	Maximum		19974.47786	
	Range		19939.35452	
	Interquartile Range		9971.389163	
	Skewness		.104	.097
	Kurtosis		-1.175	.193
	95% Confidence Interval for Mean	Lower Bound	459111.9133	
		Upper Bound	524609.9492	
LTV	Mean		491860.9313	16677.39074
	95% Confidence Interval for Mean	Lower Bound	459111.9133	
		Upper Bound	524609.9492	
	5% Trimmed Mean		458270.2508	
	Median		358998.9063	
	Variance		1.783E+11	
	Std. Deviation		422237.8085	
	Minimum		6300.978674	
	Maximum		1842200.880	
	Range		1835899.901	

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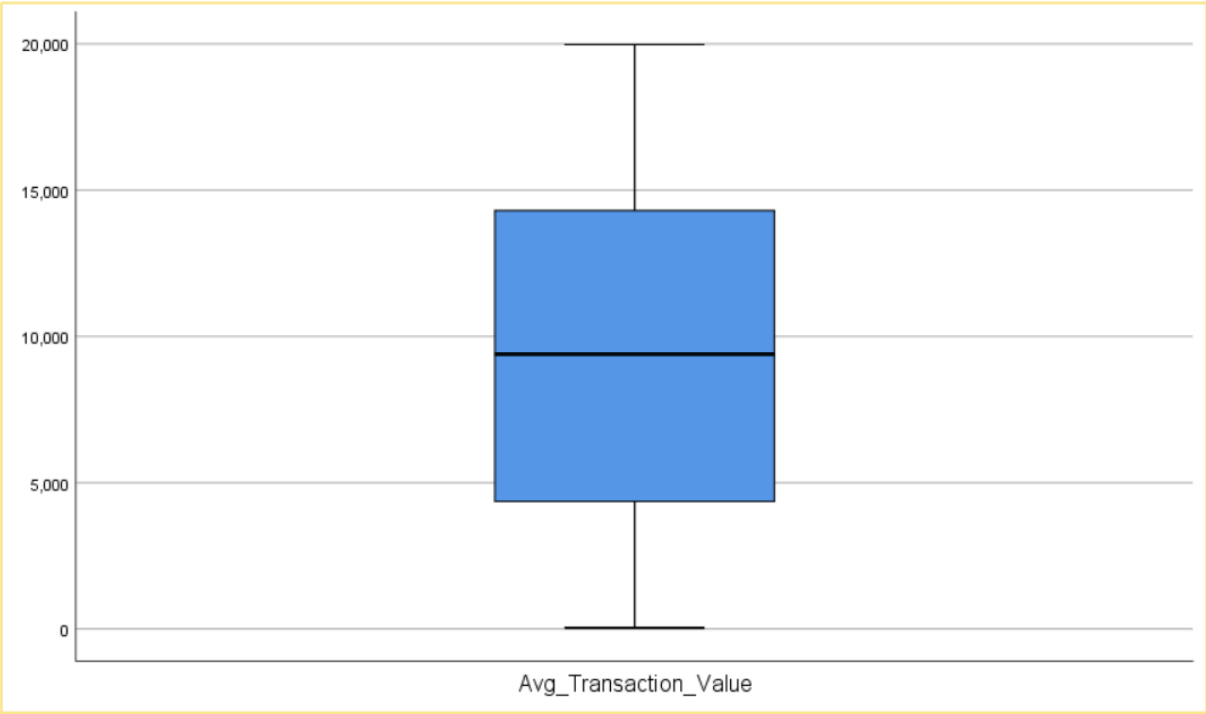
Total_Transactions



Total_Spent



Avg_Transaction_Value



LTV

